Recursive and Recursively Enumerable Languages

1. Introduction

In the Chomsky Hierarchy, languages accepted by Turing Machines (Type-0 languages) are broadly classified into two categories based on the halting behavior of the Turing Machine: **Recursive Languages** and **Recursively Enumerable (RE) Languages**.

2. Recursive Languages (Decidable Languages)

Definition: A language L over an alphabet Σ is called recursive if there exists a Turing Machine M that accepts L, and M *always halts* on every input string $w \in \Sigma^*$.

- **Behavior:**
  - If the input string $w \in L$, the TM halts and **accepts**.
  - If the input string $w \notin L$, the TM halts and **rejects**.
- **Alternative Name:** Decidable Languages (because an algorithm exists that can definitively say "yes" or "no" in a finite amount of time).
- **Closure Properties:** Recursive languages are closed under:
  - Union, Intersection, Concatenation, and Kleene Star.
  - **Complement:** (If L is recursive, its complement L' is also recursive because we can simply swap the accept and reject states of the halting TM).

3. Recursively Enumerable (RE) Languages (Partially Decidable)

Definition: A language L is recursively enumerable if there exists a Turing Machine M that accepts L, but it is not guaranteed to halt for strings that are not in L.

- **Behavior:**
  - If the input string $w \in L$, the TM halts and **accepts**.
  - If the input string $w \notin L$, the TM either halts and **rejects**, OR it enters an **infinite loop** (never halts).
- **Alternative Name:** Turing Recognizable or Partially Decidable Languages.
- **Closure Properties:** RE languages are closed under Union, Intersection, Concatenation, and Kleene Star.
  Exception: They are **NOT closed under complementation**.

4. Important Theorem connecting Recursive and RE Languages

Theorem: Let L be a language and L' be its complement. The language L is recursive if and only if both L and L' are recursively enumerable (RE).

Proof Idea: If both L and L' have Turing Machines (say M_1 and M_2) that recognize them, we can simulate both machines in parallel on an input w. Since w must belong to either L or L', one of the machines is guaranteed to halt and accept. Thus, we have a definitive algorithm that always halts, making L recursive.

5. Difference between Recursive and RE Languages

Parameters	Recursive Language	Recursively Enumerable (RE) Language
Halting property	Turing Machine always halts for all inputs.	Turing Machine halts for valid inputs, but may loop infinitely for invalid inputs.
Decidability	Decidable (Algorithm exists).	Partially Decidable (Procedure exists).
Complement	Closed under complement.	Not closed under complement.
Relationship	Every recursive language is also an RE language.	Not every RE language is a recursive language.
Example Problem	Checking if a string is a palindrome.	The Halting Problem or Post's Correspondence Problem.